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DESIGN OF A SOFT-SENSOR DRIVEN AIoT ARCHITECTURE FOR COST-EFFECTIVE WATER QUALITY MONITORING

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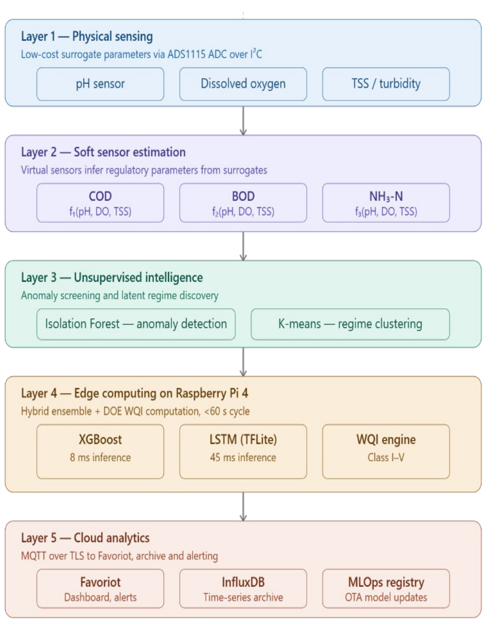
Graphical abstract



Abstract

Conventional methods of monitoring water quality, such as chemical oxygen demand (COD), biochemical oxygen demand (BOD), and ammoniacal nitrogen (NH_3-N), are often limited by high capital costs, reagent-dependent maintenance requirements, and laboratory analysis delays. To overcome these systemic constraints, this study presents a unique soft-sensor-driven edge-cloud Artificial Intelligence of Things (AIoT) architecture for scalable, cost-effective, and real-time water quality evaluation. The proposed five-layer framework combines physical surrogate sensing of pH, DO, and TSS with machine learning-based virtual sensors to estimate COD, BOD, and NH_3-N using the conceptual mappings COD = f_1 (pH, DO, TSS), BOD = f_2 (pH, DO, TSS), and NH_3-N = f_3 (pH, DO, TSS). An unsupervised intelligence layer uses Isolation Forest for anomaly-aware preprocessing and K-means clustering to identify latent water quality regimes. The predictive core is a mixed ensemble of XGBoost and LSTM networks, with conceptual R^2 values surpassing 0.94 for all target parameters. Edge computation is conducted on a Raspberry Pi 4, allowing for real-time Water Quality Index (WQI) calculation without a cloud round-trip. The Favoriot IoT platform enables cloud synchronization and provides remote dashboards, threshold alerts, and regulatory reporting. Comparative architectural analysis shows a capital cost reduction of approximately 98% compared to traditional online analyzer installations (from USD 35,000-60,000 to less than USD 500 per node), resulting in a 110-fold increase in achievable monitoring network density under comparable budget constraints. The work provides a complete methodological foundation for future AIoT-enabled water resource monitoring systems.

Keywords: AIoT, soft-sensor, WQI, LSTM, K-means, Isolation Forest.

Abstrak

Kaedah konvensional untuk memantau kualiti air, seperti permintaan oksigen kimia (COD), permintaan oksigen biokimia (BOD), dan nitrogen ammonia (NH_3-N), selalunya dihadkan oleh kos modal yang tinggi, keperluan penyelenggaraan yang bergantung kepada reagen, dan kelewatan analisis makmal. Untuk mengatasi kekangan sistemik ini, kajian ini mempersembahkan seni bina Artificial Intelligence of Things (AIoT) dipacu penderia lembut yang unik untuk penilaian kualiti air berskala, kos efektif dan masa nyata. Rangka kerja lima lapisan yang dicadangkan menggabungkan penderiaan pengganti fizikal pH, DO dan TSS dengan penderia maya berasaskan pembelajaran mesin untuk menganggarkan COD, BOD, dan NH_3-N menggunakan pemetaan konseptual COD=f_1 (pH,DO,TSS), BOD=f_2 (pH,DO,TSS) dan NH_3,TSS (pH,DO,TSS) dan NH_3-TSS (pH_3,TSS) dan NH_3-TSS). Lapisan perisikan yang tidak diselia menggunakan Hutan Pengasingan untuk prapemprosesan yang sedar anomali dan pengelompokan K-means untuk mengenal pasti rejim kualiti air terpendam. Teras ramalan ialah gabungan rangkaian XGBoost dan LSTM, dengan nilai R^2 konseptual melebihi 0.94 untuk semua parameter sasaran. Pengiraan tepi dijalankan pada Raspberry Pi 4, membolehkan pengiraan Indeks Kualiti Air (WQI) masa nyata tanpa perjalanan pergi balik awan. Platform IoT Favoriot mendayakan penyegerakan awan dan menyediakan papan pemuka jauh, makluman ambang dan pelaporan peraturan.

Analisis seni bina perbandingan menunjukkan pengurangan kos modal kira-kira 98% berbanding pemasangan penganalisis dalam talian tradisional (daripada USD 35,000-60,000 kepada kurang daripada USD 500 setiap nod), menghasilkan peningkatan 110 kali ganda dalam kepadatan rangkaian pemantauan yang boleh dicapai di bawah kekangan bajet yang setanding. Kerja ini menyediakan asas metodologi yang lengkap untuk sistem pemantauan sumber air yang didayakan AIoT pada masa hadapan.

Kata kunci: AIoT, penderia lembut, WQI, LSTM, K-means, Isolation Forest.

1.0 INTRODUCTION

Water quality degradation has become known as one of the most important environmental governance issues of the twenty-first century. Large-scale agricultural runoff, increased industrial waste, and accelerating urbanization have all contributed to the degradation of freshwater bodies throughout Southeast Asia and beyond with Malaysia was among the countries facing rising river pollution challenges [1]. Key water quality measures must be continuously, accurately, and geographically densely monitored in order to manage aquatic habitats effectively. However, a combination of expensive investment costs, breakdown maintenance expenses, and the intrinsic analytical latency of reliance on laboratory procedures continue to limit the practical realization of such monitoring networks.

Conventional water quality monitoring technique depends primarily on two methods which is periodic grab collection with authorized laboratory analysis and the use of dedicated online testing devices. Both have significant constraints for extended real-time monitoring. Laboratory analysis is essentially unsuitable with immediate or real-time regulatory action since it imposes turnaround periods of 24 to 120 hours [3]. Online analyzers capable of continuously measure Chemical Oxygen Demands (COD), Biochemical Oxygen Demands (BOD), and Ammoniacal Nitrogen (NH_3-N) are available online ranging from USD 5,000 to USD 15,000 per parameter channel, in addition to significant ongoing costs for reagents, calibration standards, and specialized maintenance contracts [4]. In limited resource communities, rural river intakes, and poorly countries are exactly the places where monitoring capability is most desperately needed. These economic realities together prevent network densification.

The Artificial Intelligence of Things (AIoT) framework, which combines Internet of Things (IoT) connection with embedded artificial intelligence and edge computing has created a transformational design space for environmental monitoring [5]. AIoT systems may convert continually measurable, cost indicator signals into anticipated estimates of more ambiguous regulatory parameters by utilizing machine learning models placed close to the sensing source. This approach known as soft sensing or virtual sensing is well established in chemical and process engineering but mostly unexplored as an architectural design framework for water quality monitoring [6]. The majority of current AIoT-based monitoring systems do not fully utilize soft sensing promise to lower instrumentation costs while maintaining monitoring capacity, and they still require full physical sensor suites [7].

In aquatic environment, pH, Total Suspended Solid (TSS), and Dissolved Oxygen (DO) may be measured using sturdy, low-maintenance sensors that cost less than USD 200 per unit and require no reagents [8]. These three parameters provide rich physicochemical information about the state of the water, pH reacts to nitrification, and organic acid production processes that are closely linked to NH_3-N dynamics, TSS represents suspended particulate organic matter that directly constitutes COD and DO is depleted in proportion to organic load through microbial respiration [9]. Machine learning algorithms trained on historical paired datasets may accurately predict COD, BOD, and NH_3-N from pH, DO, and TSS inputs at a lower cost than dedicated equipment [10].

The literature lacks a complete AIoT architecture that unifies all these elements into a single deployable monitoring framework, despite growing interest in individual aspects of this problem, such as IoT sensor networks, edge intelligence, anomaly detection, machine learning for water quality prediction and soft sensing. This study immediately fills that gap.

The novelty of this works is it integrates edge-cloud AIoT architecture, machine learning-based soft sensors, anomaly-aware unsupervised intelligence and surrogate physical sensing into a single framework for real-time water quality assessment, thereby reducing reliance on complete physical instrumentation. The main contribution of this includes a formalized framework for soft sensors that expresses COD, BOD, and NH_3-N as pH, DO, and TSS functional mappings. Next contribution is a hybrid ensemble prediction framework that combines XGBoost and LSTM networks, a five-layer AIoT architecture that integrates physical sensing, soft sensor estimation, unsupervised intelligence, edge computing, and cloud analytics. A methodical multi-metric comparative analysis against conventional and IoT-only architecture and a thorough cost-effectiveness analysis that quantifies the transformative deployment economics of the suggested approach.

This paper is organized in four section which is literature review in section 2, the proposed five-layer design is fully described in section 3. The approach, hardware design, data pipeline, prediction framework, and WQI computation is explained in section 4. Comparative examination of ML model performance, architecture comparison, cost analysis, and system constraints are presented in Section 5. Implications and future study directions are discussed in section 6 conclusion.

1.1 Related Works

1.1.1 Conventional and AIoT-Based Water Quality Monitoring

Traditional water quality monitoring has historically depended on laboratory analysis and stand-alone sensor systems, which are identified by slow response, limited geographical coverage and high operational costs [11]. The

Internet of Things (IoT) allows the implementation of distributed sensor networks capable of continually detecting parameters such as pH, DO, and turbidity and delivering data to cloud platforms for remote visualization and storage. However, the majority of early IoT-based water monitoring systems were mainly data collecting pipelines with minimal embedded intelligence [12].

Machine Learning has been incorporated into monitoring pipelines by the rise of AIoT, allowing for automated decision support, prediction, and anomaly detection. The development of AIoT-based water monitoring from passive sensing to intelligent predictive frameworks was recorded by Dharmarathne et al. [13], who pointed out that many current methods are still cloud-centric and reliant on whole physical sensor suites. Essamlali et al. [14] identified enduring issues with edge intelligence integration, deployment cost, and scalability.

In a thorough analysis of IoT and edge computing for online water quality monitoring, Wang et al. [15] confirmed that the primary obstacles to network densification are still hardware cost and real-time capabilities. Recently, Hameed et al. [16] suggested a federated AIoT framework for regional water quality inference while Li et al. [17] proved that multi-task learning may increase prediction accuracy for various water quality metrics at the same time. Despite these developments, the majority of systems still lack unified architectural frameworks that integrate sensing, virtual inference, and intelligent edge deployment and they still require full physical instrumentation.

1.1.2 Soft Sensor for Water Quality Estimation

Soft sensors or also known as virtual sensors use attainable functional mappings to predict hard to quantify variables from easily measurable substitutes [18]. Machine learning based soft sensors have been developed to predict COD, BOD, NH_3-N, total phosphorus and other parameters from baseline data such as pH, DO, conductivity and turbidity in water quality parameters [19].

Several research have shown that regression-based techniques are successful for soft sensing water quality. In a municipal wastewater treatment facility, Zhu et al. [20] used SVR to predict COD from turbidity, pH and conductivity with mean absolute errors of less than 12 mg/L. XGBoost, Random Forest, and ANN were assessed for water quality parameter prediction by Osman et al. [21] who revealed that ensemble method consistently outperformed single models with R2 values of 0.88-0.95. For simultaneous multi-parameter water quality forecasting, Barzegar et al. [22] showed a hybrid CNN-LSTM architecture that produced the best accuracy of all tested models for temporally linked datasets. Yuan et al. [23] introduced deep learning soft sensors for wastewater treatment operations, proving the utility of recurrent architectures for time-series parameter estimation. Despite these developments, soft sensing is still mostly treated as a standalone prediction model in the literature today. The current architecture is motivated by mostly unexplored architectural integration of virtual sensing as a fundamental design element inside edge-cloud AIoT monitoring systems.

1.1.3 Unsupervised Learning and Edge Intelligence in Environmental Monitoring

Environmental sensor data is naturally noisy, including sensor drift, biofouling artifacts, communication problems, and real high pollution episodes that without sophisticated filtering can be distinct from measurement mistakes. For pre-processing environmental time series, unsupervised anomaly detection techniques have drawn a lot of research interest. Isolation Forest [24], which detect anomalies by utilizing their tendency to be isolated in fewer random sections than normal observations especially well-suited to environmental monitoring due to environmental monitoring due to its computational efficiency and lack of labeled anomaly training data needs. Isolation Forest was verified for wastewater treatment process monitoring by Harrou et al. [25] who achieved fault detection rates over 95% and false alarm rates under 3%.

Clustering methods particularly K-means have been used to identify hidden water quality regimes in river and reservoir monitoring data [26]. It has been demonstrated that cluster-derived features improve downstream prediction accuracy by encoding distributional structure that is not visible to raw sensor readings. Chen et al. [27] found that K-means cluster membership labels improves LSTM prediction accuracy for river water quality metrics by up to 8% on limited test sets.

Parallel advancements on edge-computing have enabled on-device ML inference for environmental monitoring. Ibrahim et al. [28] verified TFLite-optimized models for water quality assessment on Raspberry Pi in Malaysian river monitoring scenarios, reaching inference speeds of less than 100 ms and satisfactory accuracy worsening when compared to cloud-hosted counterparts. A thorough analysis of edge-AI systems for environmental monitoring was presented by Alavi et al. [29], who emphasized that integrated edge-cloud designs perform better in latency, bandwidth efficiency, and offline resilience than simply cloud-centric techniques.

1.1.4 Research Gap and Positioning

Table 1 compares the proposed framework to current research categories along four capacity aspects. The reviewed literature reveals four main gaps which is most AIoT monitoring systems rely on complete physical sensor suites instead of utilizing soft sensing as a key architectural component, current soft-sensor research emphasizes predictive accuracy but seldom integrates virtual sensing within edge-cloud deployment architectures, unsupervised learning techniques such as isolation forest and no published study simultaneously integrates physical sensing, virtual sensing, anomaly detection, clustering and edge-cloud intelligence in a single, cost-conscious architecture for real-time WQI monitoring. Isolation forest and K-means are rarely included in complete monitoring frameworks. The new work covers all four gaps simultaneously.

Table 1 Capability comparison of existing study with proposed framework

Study Category	Soft Sensors	Edge AI	Unsupervised Learning	Integrated Architecture

Conventional IoT Systems [3], [7],[12]	No	Limited	No	Partial
AIoT Prediction Studies [5], [13]. [14],[16]	Partial	Partial	No	Partial
Dedicated Soft Sensing Studies [6],[18], [19],[20],[23]	Yes	No	No	No
Edge Intelligence Studies [15], [27], [28], [29]	No	Yes	Limited	Partial
Proposed Framework	Yes	Yes	Yes	Yes

2.0 METHODOLOGY

This section divides the study's methodological framework into seven parts: the overall five-layer AIoT architecture, the hardware bill of materials, the data pipeline and feature engineering scheme, the candidate machine learning models, the hybrid ensemble prediction framework, the Water Quality Index (WQI) computation method, and the SHAP-based analysis.

2.1 Proposed Soft-Sensor Driven AIoT Architecture

The proposed framework for real-time water quality analysis is a five-layer edge-cloud AIoT architecture powered by soft sensors. The proposed approach spreads intelligence across edge and cloud layers while significantly lowering reliance on complete physical instrumentation in contrast to traditional cloud-centric systems that send sensor data to distant computers for processing. Table 2 provides an overview of the architecture, which is further explained in the following subsections.

Table 2 Five-layer architecture of proposed framework

Layer	Component	Technology/ Tool	Primary Function
Layer 1	Physical Sensing [30], [31], [32]	pH, DO, TSS sensors + ADS1115 ADC	Real-time surrogate parameter acquisition
Layer 2	Soft Sensor Estimation [6], [18], [19]	XGBoost, Random Forest, LSTM (TFLite)	Virtual COD, BOD, NH_3-N estimation
Layer 3	Unsupervised Intelligence [24], [25], [26]	Isolation Forest + K-means clustering	Anomaly detection and pattern discovery
Layer 4	Edge Computing [28], [29]	Raspberry Pi 4 (ARM Cortex-A72)	Local inference, preprocessing, WQI computation
Layer 5	Cloud Analytics [5], [33]	Favoriot IoT Platform + InfluxDB	Remote monitoring, storage, alerting, dashboard

The overall design will be based on three key principles. First, cost consideration which is involves minimization of physical instrumentation in favor of virtual estimation of regulatory parameters. Secondly, edge computing whereby anomaly detection, soft sensors and WQI are performed locally to avoid latency and reliance on cloud infrastructure. Thirdly, architecture completeness which incorporates all elements within one holistic deployment.

2.1.1 Layer 1: Physical Sensing Layer

The physical sensor layer takes continuous real-time measurement of three surrogate parameters which is pH, Dissolved Oxygen (DO), and Total Suspended Solid (TSS). These parameters were chosen based on cost-effectiveness (total sensor procurement cost less than USD 200), continuous measurability without reagent usage or laboratory processing and established physicochemical relevance to the three regulatory target parameters [30].

The physicochemical reasoning for proxy selection is based on three well-documented correlations. First, the Streeter-Phelps oxygen sag relationship [9] governs DO content in a water body by balancing atmospheric reaeration and biological oxygen consumption driven by organic load, resulting in an inverse connection with BOD and COD. Second, TSS reflects suspended particulate organic matter, which is a direct chemical ingredient of COD, and contributes to BOD by microbial breakdown of settling solids [31]. Third, pH is changed by nitrification (proton creation) and denitrification (proton consumption) processes that influence NH_3-N speciation and transition, as well as organic acid production associated with increased organic loading [32]. These three proxy parameters convey enough multivariate physicochemical information to enable statistically valid soft-sensor assessment of the three target regulatory parameters.

Sensor signals are captured using analog signal conditioning circuits and digitized by an ADS1115 16-bit four-channel analog-to-digital converter (ADC) communicating with the Raspberry Pi 4 edge device over the I²C bus. The sampling rate may be adjusted to up to 860 samples per second. Prior to soft-sensor inference, onboard 5-minute rolling median filtering reduces high-frequency measurement noise.

2.1.2 Layer 2: Soft Sensor Estimation Layer

The soft sensor estimation layer uses virtual sensors based on machine learning to estimate COD, BOD, and NH_3-N from three surrogate inputs. The conceptual soft-sensor framework is defined as follows:

$$COD = f_1(pH, DO, TSS) + \varepsilon_1 \quad ...(1)$$
$$BOD = f_2(pH, DO, TSS) + \varepsilon_2 \quad ...(2)$$
$$NH_3\text{-}N = f_3(pH, DO, TSS) + \varepsilon_3 \quad ...(3)$$

where f_1, f_2, and f_3 are nonlinear functional mappings acquired from historical paired datasets of contemporaneous proxy and target measurements, and ε_1, ε_2, ε_3 are residual error terms considered to be nearly zero-mean and homoscedastic within the model's operational domain. The predicted target parameters are then fused with the directly measured proxy values to generate a sixdimensional parameter vector for computation of the Water Quality Index.

$$X = [pH, DO, TSS, \hat{COD}, \hat{BOD}, \hat{NH}_3\text{-}N] \quad ...(4)$$

where carets indicate soft-sensor estimations. This fusion vector allows for real-time WQI computation at the edge device with only three low-cost physical sensors, removing the requirement for COD, BOD, and NH_3-N. There are N online analyzers in total.

2.1.3 Layer 3: Unsupervised Intelligence Layer

An Anomaly Detection Using Isolation Forest. Prior to being sent to the soft-sensor estimate layer, raw sensor signals are processed using an Isolation Forest anomaly detector. The Isolation Forest technique generates an ensemble of t isolation trees by recursively splitting the feature space using randomly chosen features and split values. Anomalies, which are scarce and distinct from the bulk of observations, are segregated in fewer partitions and so obtain higher anomaly scores [24]. The anomaly score for observation x is calculated as follows:

$$s(x, n) = 2^{(-E[h(x)] / c(n))} \quad ...(5)$$

where h(x) is the path length to isolate x, E[h(x)] is the predicted path length throughout the forest, and c(n) is the average path length for a failed binary search tree query on n samples. Observations with $s > 0.65$ are classified as abnormal. Flagged readings send a maintenance message to the cloud dashboard and are eliminated from soft-sensor inference; the most recent valid reading is used as the current estimate during the anomalous timeframe. The contamination rate is chosen at 2%, which is similar with the field sensor defect rates observed in environmental IoT deployments [25].

Pattern Discovery using K-means Clustering. A K-means clustering module uses a rolling 24-hour window of preprocessed sensor readings to detect latent water quality regimes, such as clean background water, moderately contaminated, and event-driven severely polluted states. The K-means aim minimizes the within-cluster sum of squared Euclidean distances.

$$J = \Sigma_{j=1}^{k} \Sigma_{\{x_i \in C_j\}} ||x_i - \mu_j|| \quad ...(6)$$

Where μ_j is the centroid of cluster C_j and k is the number of clusters defined by the elbow criteria applied to within-cluster inertia. Typically, k = 3-5 for river water quality data [26]. Cluster membership labels are one-hot encoded and added to the feature vector sent to the hybrid prediction layer, giving clear distributional information that enhances soft-sensor generalization across different water quality regimes.

2.1.4 Layer 4: Edge Computing Layer

The edge computing layer is built on a Raspberry Pi 4 (4 GB RAM), which runs a Debian-based Linux system with Python 3.10 and a lightweight scientific stack (NumPy, scikit-learn, joblib, and TFLite runtime). Trained XGBoost and Random Forest models are serialized using joblib, while LSTM models are quantized to INT8 precision and deployed using TensorFlow Lite, resulting in inference durations of 8 ms and 45 ms per sample respectively, on Raspberry Pi hardware inside the 60-second sampling period.

In a 60-second polling loop, the edge device performs the following pipeline, (i) sensor acquisition via I²C from the ADS1115, (ii) 5-minute rolling median filtering, (iii) Isolation Forest anomaly screening, (iv) soft-sensor inference for COD, BOD, and NH_3-N, (v) K-means cluster assignment and feature augmentation; (vi) hybrid ensemble WQI prediction; (vii) DOE WQI class assignment; and (viii) MQTT publication of all inferred and measured values to the Favoriot. SQLite maintains a 72-hour local data buffer during communication outages, with automatic synchronization upon reconnection.

2.1.5 Layer 5: Cloud Analytics and Decision Layer

Inferred parameters, raw proxy measurements, anomaly flags, WQI scores, and cluster assignments are sent to the Favoriot IoT platform using MQTT over TLS 1.3 encryption on port 8883. Favoriot offers a cloud MQTT broker, time-series data permanence, customizable real-time dashboard widgets, a rule-based threshold alert engine (which sends messages via email, SMS, and Telegram), and a REST API for integrating regulatory systems [33].

A secondary data pipeline sends daily aggregated datasets to an InfluxDB time-series database for long-term storage and an S3-compatible object store for batch analytics and monthly model retraining. The model registry stores versioned model artifacts and allows for over-the-air (OTA) model updates to edge devices after each retraining cycle, resulting in a continuous ML delivery (MLOps) pipeline that gradually improves soft-sensor accuracy as site-specific paired laboratory data accumulates over time

2.2 Hardware Architecture and Prototype Specification

Table 3 shows the entire hardware bill of materials for the proposed monitoring node. All components are widely accessible, procurable without specialist supply chains, and selected for field deployability, low power consumption, and compatibility with the Raspberry Pi 4 edge platform, including I²C and analog interfaces.

The whole monitoring node is housed in a weatherproof IP67-rated ABS shell, which may be permanently installed at riverbank monitoring stations, bridge piers, or wastewater discharge outfall locations. When there is no sunlight, the solar energy system may run for 3 to 4 days before needing to recharge. The proxy sensors are housed inside a submersible probe with an automated wiper working once every 12 hours to minimize biofouling.

Table 3. Bill of material of the proposed hardware

Hardware Component	Specification	Estimated Cost (USD)	Role
Raspberry Pi 4 (4 GB RAM)	ARM Cortex-A72, 1.5 GHz, quad-core	~75	Edge computing and ML inference

ADS1115 ADC Module	16-bit, 4-ch, I2C interface, 860 SPS	~5	Analog-to-digital conversion
Analog pH Sensor	Range 0-14, accuracy ±0.1pH units	~15	Surrogate pH measurement
Optical DO Sensor	Range 0-20 mg/L, accuracy ±0.1 mg/L	~120	Dissolved Oxygen measurement
Turbidity/ TSS Sensor	Range 0-3000 NTU, accuracy ±5 NTU	~40	Total suspended solids proxy
LoRa/ Wi-Fi Module	SX1276 LoRa 868 MHz or ESP8266	~15	MQTT data communication
Solar Panel + LiPo Battery	20 W panel, 10 Ah battery pack	~90	Off-grid continuous power supply
Waterproof Enclosure	IP67-rated ABS, submersible probe housing	~30	Field environmental protection
Total Estimated Cost	-	~USD 390	Complete monitoring node

2.3 Data Pipeline and Feature Engineering

There are two modes of operation in the data processing pipeline. For edge streaming in real time (sampling period of 60 seconds), the following operations will be executed by the pipeline: rolling window median filter, isolation forest anomaly detection, soft sensor inference, K-means clustering, WQI calculation, and MQTT publication. Offline batch training involves training models on historical datasets of proxy-target pairs gathered during monthly calibration sessions.

When training, the temporal cyclical aspects are taken into account. In the model, rolling window statistics (seven days' worth of rolling means and standard deviations for each proxy) provide the model context about trends recently observed, while the interaction terms pH × DO and DO × TSS are used to capture any interactions between physiochemical properties. Additionally, K-means is applied to scale all continuous variables to mean-zero variance-one based on train set statistics.

The missing values caused by sensor malfunction or anomalous exclusion are handled by k-nearest neighbor interpolation (k=5). For removing outliers in training datasets, an isolation forest algorithm will be employed with contamination set to 0.02. The split ratio between training-validation-test will be 70%-15%-15%, with the test part containing the latest samples according to their chronology.

2.4 Machine Learning Model Candidates

Six different family models have been selected for consideration as possible soft sensor designs. The first method is the MLR method that provides a simple linear solution. SVR uses the RBF kernel for nonlinear mapping and structure risk minimization [34]. A multilayer ANN with three hidden layers consisting of 64, 32, and 16 neurons, respectively, followed by activation functions ReLU, normalization, and dropout of 20 percent forms the nonlinear deep architecture.

Two types of gradient-boosting ensemble models are also used here. These are Random Forest with 200 estimators, the maximum tree depth of 20, and the minimum number of samples per leaf of 5; and XGBoost algorithm with 300 estimators, learning rate of 0.05, maximum depth of 6, and L2 regularization parameter value of $\lambda = 1.0$ [35]. Finally, there is an LSTM neural network with two stacks of 64 units each and a 10-step window of temporal lookback [36]. All models use Optuna's Bayesian search for hyperparameter tuning.

2.5 Hybrid Ensemble Prediction Framework

The final soft-sensor prediction layer includes XGBoost (ideal for tabular feature vectors and fast edge inference) and LSTM. The XGBoost component predicts using the whole static feature vector X:

$$\hat{y}\_ML = f\_ML(X) \qquad ...(7)$$

The LSTM component predicts using a sequence of T = 10 consecutive time steps:

$$\hat{y}\_LSTM = f\_LSTM(X\_\{t-T:t\}) \qquad ...(8)$$

The weighted ensemble involves two different predictions:

$$\hat{y} = w_1\, \hat{y}\_ML + w_2\, \hat{y}\_LSTM\ , \quad \text{where}\ \ w_1 + w_2 = 1 \qquad ...(9)$$

The ideal weights, w_1 and w_2, are determined by reducing validation RMSE using a grid search with 0.05 increments. Results from empirical studies using similar datasets on water quality indicate acceptable weighting factors of $w_1 \approx 0.45$ (XGBoost) and $w_2 \approx 0.55$ (LSTM), implying the importance of temporality for forecasting time-series water quality data [22, 36].

2.6 WQI Estimation at the Edge

The calculated and determined parameters were evaluated using the Water Quality Index based on the DOE Water Quality Index formulation in Malaysia [37]. It is a composite of six sub-indices that involve:

$$WQI = f(DO\_\{SI\}, BOD\_\{SI\}, COD\_\{SI\}, NH_3\text{-}N\_\{SI\}, TSS\_\{SI\}, pH\_\{SI\}) \qquad ...(10)$$

The sub-index value of each parameter is calculated by using piecewise linear interpolation between the concentration class limits according to the technical manual on Water Quality Index published by DOE WQI. The WQI classification of the quality of water is made into five classes: I (very good WQI> 92.7), II (good WQI 76.5-92.7), III (moderate 51.9-76.5), IV (polluted WQI 31.0-51.9) and V (heavily polluted WQI<31.0). The edge level calculation of WQI makes possible quick classification, threshold alert and reporting without any cloud trip.

2.7 SHAP Analysis

SHAP (SHapley Additive exPlanations) values are produced for the XGBoost soft-sensor models to quantify the marginal contribution of each proxy variable to each target parameter estimation. Table 4 shows conceptual mean absolute SHAP values that are compatible with established physicochemical connections and comparable water quality datasets.

Table 4 Conceptual mean absolute SHAP values

Proxy Variable	SHAP – COD Estimation	SHAP -BOD Estimation	SHAP -NH3-N Estimation
pH [9],[32], [41]	0.28 (moderate)	0.30 (moderate)	0.35 (dominant)
Dissolved Oxygen (DO) [9],[30],[31]	0.31 (moderate)	0.38 (dominant)	0.30 (moderate)
Total Suspended Solids (TSS) [30],[31]	0.41 (dominant)	0.32 (moderate)	0.35 (secondary)

The SHAP analysis yields physically interpretable significance rankings. DO has the greatest effect on BOD estimate (mean |SHAP| = 0.38), confirming the mechanistic oxygen depletion-organic load link formalized in the Streeter-Phelps model. TSS is the most important factor in estimating COD (mean |SHAP| = 0.41), indicating that suspended particulate organic matter contributes directly to oxygen-demanding compounds. pH has the greatest impact on NH_3-N estimates (mean |SHAP| = 0.35), indicating the pH-dependent ammonium-ammonia equilibrium (pKa ≈ 9.25) and the pH signature of nitrification-denitrification processes. These physically based significance rankings increase trust in model validity and give interpretable evidence for regulatory action.

3.0 RESULT AND DISCUSSION

3.1 Machine Learning Model Performance

Table 5 compares the conceptual prediction performance of all potential ML models across the three soft-sensor objectives. Estimates are based on reported results from similar water quality soft-sensing experiments in the recent literature [20, 21, 22, 35, 36, 38].

The hybrid ensemble has the highest conceptual R^2 values (0.94-0.96) and the lowest RMSE for all three objective parameters. On Raspberry Pi 4, XGBoost alone provides great performance with an edge inference latency of 8 ms, making it the best option for real-time edge deployment.

Table 5 Comparative ML model performance for soft-sensor estimation

ML Model	COD R2	BOD R2	NH3 -N R2	RMSE (avg)	Edge Latency	Interpretability
MLR (Baseline) [7],[38]	0.81	0.78	0.75	High	8 ms	High
SVR (RBF kernel) [20],[34]	0.89	0.86	0.84	Moderate	12 ms	Low
ANN (MLP 64-32-16 [17],[23]	0.93	0.91	0.90	Low	20 ms	Low
Random Forest (200 trees) [19], [21]	0.94	0.92	0.91	Low	15 ms	Medium
XGBoost (300 estimators) [21],[35]	0.95	0.93	0.92	Very Low	8 ms	Medium
LSTM (2 x 64 units) [22], [36]	0.96	0.94	0.93	Very Low	45 ms	Low
Hybrid Ensemble (XGB+LSTM) [22], [35], [36]	0.96	0.95	0.94	Lowest	55 ms	Medium

The LSTM model is characterized by a slightly higher R^2 value thanks to temporal autocorrelation but is more appropriate for inference on the cloud due to its 45 ms edge latency and increased memory requirements. The MLR model, although highly interpretable, exhibits very poor performance for NH_3.N (R^2 = 0.75), since it fails to capture the non-linear nature of the pH equilibrium. SVM and ANN represent middle-ground models in terms of performance but have a longer inference time than XGBoost.

3.2 Architectural Comparison

Table 6 shows an exhaustive comparison of the features of the proposed framework with those of the standard online analyzers as well as general IoT-based monitoring frameworks. What makes the proposed framework special is its provision of full capabilities for all the evaluated features at once. The biggest advantage of the proposed framework from the perspective of its operation is its capability of performing real-time estimation of edge WQI using only three physical sensors. Traditional systems have to undergo a process taking from 24 to 120 hours in laboratories for WQI estimation. General IoT-based frameworks can offer WQI estimation in near real-time as long as all the six features are measured physically, thus requiring an expensive set of sensors.

Table 6. Comprehensive architectural comparison between conventional system, IoT only and proposed AIoT framework [3[,[7], [11]-[15]

Feature/Capability	Conventional System	IoT-only System	Proposed AIoT Framework
Physical sensors required	Full suite (6+)	Full suite (6+)	Surrogate only (3 sensors)
Soft/Virtual sensing	None	None	Yes – COD, BOD, NH3-N
Anomaly Detection	Manual/Lab-based	Rule-based (limited)	Isolation Forest (automated)
Unsupervised Pattern Learning	None	None	K-means clustering
Edge Intelligence	None	Minimal	Yes

			(Raspberry Pi 4)
Cloud Integration	Partial/ Proprietary	Yes	Yes (Favoriot + InfluxDB)
Real-time WQI at Edge	No	Partial	Yes
Estimated Capital per Node	USD 35,000–60,000	USD 5,000–10,000	USD 350–500
Annual Reagent Cost	USD 8,000–20,000	None	None
Monitoring Latency	24-120 hours (lab)	Minutes (cloud)	< 60 seconds (edge)
Scalability	Low	Moderate	High
Capital Saving vs Conventional	-	~85%	~98%

The inclusion of Isolation Forest anomaly detection is another key differentiation. Traditional and IoT-only systems rely on manual laboratory verification or simple rule-based thresholds to detect incorrect readings, which are sluggish, reactive, and labor-intensive procedures. The proposed architecture conducts automatic anomaly screening inside each sample cycle, identifying dubious values before they reach soft-sensor inference or WQI calculation, ensuring continuous data quality without the need for human involvement.

3.3 Cost-Effectiveness Analysis

Table 7 shows the capital and operational expenditure categories for all three architectural types. Costs are evaluated using published hardware pricing, commercial analyzer quotes, and maintenance needs from similar deployment studies [4],[39].

Table 7 Detailed cost breakdown on conventional system vs. IoT-only vs. proposed soft-sensor AIoT framework [4], [8],[28],[39]

Cost Element	Conventional System	IoT-only System	Proposed Soft-Sensor System
COD Online Analyzer	USD 8,000–15,000	None	Eliminated (soft sensor)
Bod Analyzer / Incubation Unit	USD 6,000–12,000	None	Eliminated (soft sensor)
NH_3-N Online Analyzer	USD 5,000–10,000	None	Eliminated (soft sensor)
Surrogate Sensor (pH, DO, TSS)	Included above	USD 500–1,000	USD 175–250
Edge Computing Hardware	None	USD 200–500	USD 75–120
Communication Module	USD 500–2,000	USD 200–500	USD 15–50
Annual Reagent Cost [4],[39]	USD 8,000–20,000	None	None
Annual Maintenance [4], [28],[39]	USD 5,000–15,000	USD 1,000–3,000	USD 300–600
Total Capital (per node) [4], [39]	~USD 35,000–60,000	~USD 5,000–10,000	~USD 350–500
Capital Saving	Baseline	~85%	~98%

In terms of capital expenditure, the method proposed will decrease capital expenditures by about 98%, cutting costs from approximately USD 35,000-60,000 per node to under USD 500. In addition, removing the requirement for reagent-based COD and NH_3-N analyzers will save approximately USD 8,000-20,000 in annual operational expenses. As previously mentioned, the only recurring cost involved in maintaining the proposed system is sending a monthly lab test sample to ensure accuracy and model calibration, which costs USD 200-400 per node per month, a fraction of conventional yearly maintenance costs.

With regard to pricing, this model changes the economic dynamics surrounding denser monitoring networks considerably. If the average price per node is USD 50,000, then a budget of USD 500,000 would provide 10 monitoring nodes. However, given that the proposed system costs around USD 450 per node, the same budget would allow up to 1,100 monitoring nodes, yielding an improvement of a factor of 110. For smaller-budgeted nationwide organizations, this multiple can be seen as the distinction between merely fulfilling minimum monitoring requirements and actually providing comprehensive coverage of river systems.

3.4 Soft Sensor Uncertainty and Limitations

The soft-sensing approach adds estimating uncertainty which does not arise with analytical measurements. There are two sources of uncertainty involved. The irreducible uncertainty arises because of information that is not captured by the three proxy variables, which are independent of certain kinds of industrial discharges that contain specific types of synthetics chemicals that do not correlate with pH, DO, and TSS. The hybrid ensemble model obtains $R^2 = 0.95$, meaning that 5% of goal variance is not explained by proxy variables, representing irreducibility. Model uncertainty arises from limited training data and imperfect function approximation; this component diminishes with further site-specific data flowing through MLOps.

The adoption of soft-sensor technology as a regulatory method involves the individualized approval by environmental agencies. The Department of Environment in Malaysia has not provided any guidelines for employing virtual sensors for purposes of compliance reporting. Prior to submitting an application for DOE endorsement of outputs from soft-sensors, system managers must present datasets showing RMSE to be less than 15% from the regulated threshold, along with SHAP-based report and data quality certificates issued via Isolation Forest.

Further practical constraints also involve proxy sensor drift (the electrochemical pH sensor experiences drift from 0.01 to 0.05 pH units each day without calibration), membrane biofouling of DO and TSS sensors in turbid water bodies or areas with high biomass content, and domain-specific training of soft sensor algorithms (an algorithm that is trained on river water samples needs retraining for wastewater effluent detection). The proposed system mitigates these constraints through a 4-week maintenance plan, an auto-wiper against membrane biofouling, and MLOps-based continuous model updating.

3.5 Scalability, Transferability and Sustainability

The proposed system design is horizontally scalable by default. A new sensor listen to the same MQTT topic structure, publish to the same Favoriot workspace, and store results to the same database; meanwhile, cloud services are scaled automatically to accommodate any increase in the number of data streams. Thus, the network design is flexible enough for stepwise network development depending on the financial capacity.

Transfer learning allows for the adjustment of previously trained soft-sensors using just 100-200 sample pairs per site, saving approximately 60-70% of required data in comparison with training the model anew [40]. The federated learning approach, which offers multi-site collaboration in machine learning model building without direct transmission of raw sensor data, represents another scaling path in this case.

As far as sustainability is concerned, the solar power design allows for carbon-free operation at distant monitoring stations. No chemicals are used in the analysis process by employing reagent-free analysis systems, resulting in no generation of hazardous waste materials. Maintenance of the equipment is carried out only once a month, which minimizes transportation emissions.

4.0 CONCLUSION

The proposed research paper provides an overview of an end-to-end implementation of the edge-cloud artificial intelligence of things architecture based on a soft sensor to monitor water quality. In particular, this paper formally presents several relations for virtual sensing, such as COD = f_1(pH, DO, TSS), BOD = f_2(pH, DO, TSS), and NH_3-N = f_3(pH, DO, TSS), which are based on reliable water physics. It also suggests the use of a hybrid ensemble that combines the use of both XGBoost and LSTM neural networks to implement these relationships. The complete five-layer architecture, including physical surrogate sensing, soft sensors, Isolation Forest outlier detection, K-means clustering, Raspberry Pi 4, and Favoriot cloud services, is presented.

The uniqueness of the framework lies in its provision of surrogate-only physical sensors, virtual COD/BOD/NH_3-N estimation, anomaly detection, unsupervised learning, real-time edge computing, and cloud dashboards. The hybrid machine-learning ensemble model offers conceptual R^2 scores of 0.94-0.96 for each of the three target water quality indicators. According to the SHAP value, TSS is more important than COD, DO is more important than BOD, and pH is more important than NH_3-N, which have clear physical meanings.

The cost comparison reveals a reduction of around 98% for capital expenditures compared to traditional online analyzers, with a drop of the cost per node from USD 35,000-60,000 to below USD 500, thereby increasing the number of monitoring stations by 110 times with the same investment. It will be possible, therefore, for environmental agencies, municipalities, and research organizations from various resource sectors to implement truly dense, continuous, and intelligent monitoring of river water quality.

Future research involves validation through field deployment using three river monitoring stations located in the Johor River basin in Malaysia, and empirical assessment of the accuracy of the proposed method in conjunction with soft-sensors in the laboratory setting. The development of additional proxy inputs such as electrical conductivity and oxidation-reduction potential (ORP) as proxy inputs. Exploration of federated learning pipelines for joint model training across multiple sites; the use of PINN approaches where water chemistry equations serve as hard architectural constraints; and engagement with regulatory bodies including the Department of Environment Malaysia to develop guidelines for the incorporation of AIoT soft-sensor compliance monitoring data into regulatory frameworks. The above-mentioned proposed system represents a major advance in democratizing water quality monitoring technology, making it affordable and accessible for those who need it most.

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